# XENONnT Ionization Data Probe Solar Chameleons

*First search for solar chameleons using low-energy S2-only data extends sensitivity to sub-keV recoils.*

**arXiv:2609.33198; XENONnT ionization-only data; solar chameleons; screened dark energy.**

Authors: Guan-Wen Yuan

arXiv:2609.33198 doi:10.48550/arXiv.2609.33198

## Background

Screened scalar fields link dark energy to laboratory-scale new physics while evading fifth-force constraints. Solar production offers a direct probe, and ionization-only measurements extend liquid-xenon sensitivity to sub-keV energies.

> **Previous electronic-recoil analyses lacked sensitivity below ~0.1 keV, leaving low-energy chameleon signatures unexplored.**

## What this paper is selling

This work performs the first solar chameleon search using XENONnT ionization-only data, reaching ~0.04 keV. It establishes low-energy ionization as a new channel for testing screened dark-energy models.

## Key results to remember

- No significant excess observed; 95% C.L. upper limit log10 βeff < -3.64.
- Constraint weaker than previous electronic-recoil analysis due to rapidly falling disformal rate.
- Low-energy ionization data open a new experimental channel for solar chameleons.

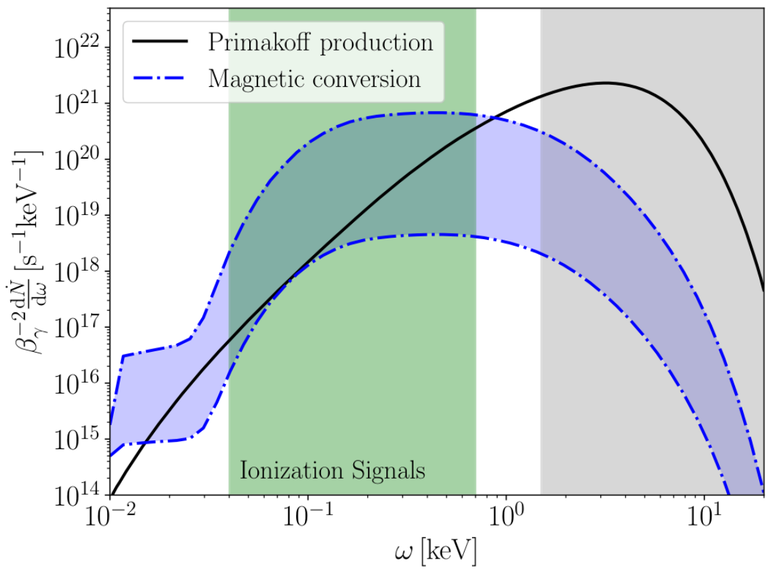



**Figure 1 |** Solar chameleon production spectrum shows Primakoff and magnetic conversion contributions, with the green band marking the electronic-recoil interval probed by XENONnT.

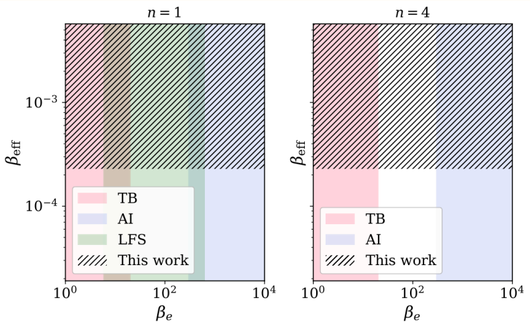



**Figure 6 |** Constraints in the βeff–βe plane for n=1 and n=4, with the hatched region excluded at 95% C.L.